

PROJECT PROPOSAL

HeartLens AI

Edge ECG Monitoring with On-Device AI Inference

Field	Details
Concept	Offline-capable ECG screening device with edge AI inference
Target Cost	~\$12 USD (full bill of materials per unit)
Connectivity	None required — fully offline operation
Device Class	Preventative screening aid (non-diagnostic)
MCU Platform	ESP32 (dual-core, 240 MHz, 520 KB RAM)
AI Models	LSTM denoiser + 1D-CNN classifier (TFLite Micro, int8)
Detects	AFib, PVC, Tachycardia, Bradycardia, ST abnormality, Normal
Build Timeline	16 weeks (part-time)
Open Source	All hardware, firmware, models, and documentation
Publication Target	Sensors (MDPI, Q2) / IEEE IoT Journal

"Every year, millions of people suffer preventable heart attacks not because treatment does not exist, but because warning signs went unnoticed until too late. HeartLens AI is built to close that gap — for anyone, anywhere, at a cost lower than a restaurant meal."

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01 Motivation: The Problem We Are Solving

Background

Cardiovascular disease is the leading cause of death globally, responsible for approximately 18 million deaths per year. What makes this statistic particularly painful is that a large proportion of serious cardiac events — heart attacks, strokes, sudden cardiac arrest — are preceded by detectable warning signs that appear weeks or months in advance.

Conditions like atrial fibrillation (AFib), abnormal heart rhythms (arrhythmias), and ST-segment irregularities show up clearly in an electrocardiogram (ECG). Catching them early means a doctor can intervene with medication, lifestyle changes, or monitoring — before a manageable condition becomes a life-threatening emergency.

CORE PROBLEM

The technology to detect these warning signs has existed for decades. The problem is access. A clinical ECG requires a hospital visit, specialist equipment, and trained staff to interpret results. For billions of people in rural areas, low-income settings, or healthcare-underserved regions, that visit simply does not happen — until something goes wrong.

Gap in Existing Solutions

Existing consumer cardiac devices (AliveCor KardiaMobile, Withings ScanWatch) cost \$100–\$400, require a smartphone, and depend on cloud connectivity for AI analysis. They were designed for the worried-well in wealthy urban markets.

They were not designed for a 58-year-old farmer in a village without reliable internet, or a night-shift factory worker who has never had reason to think about his heart rhythm. HeartLens AI is designed for those people.

The goal is not to replace a cardiologist. The goal is to be the thing that tells someone: 'You should go see a cardiologist' — before they end up in an ambulance.

02 Project Goals & Objectives

Primary Goal

HeartLens AI aims to build, validate, and document a fully offline, sub-\$15 ECG monitoring device that runs AI inference directly on the hardware and outputs plain-language risk assessments — no smartphone, no internet, no medical training required.

PRIMARY OBJECTIVE

Demonstrate that edge AI inference on a \$3.50 microcontroller can provide meaningful cardiac screening signal — reliably distinguishing normal rhythms from patterns that warrant medical follow-up — in real-world noisy conditions.

Secondary Goals

- Produce a rigorous noise robustness study comparing real-world vs. synthetic ECG noise conditions
- Open-source all hardware designs, firmware, trained models, and documentation
- Demonstrate deployment viability through a small real-world feasibility study
- Contribute a publishable research artifact to the IoT health monitoring literature
- Target publication in Sensors (MDPI, Q2) or IEEE Internet of Things Journal

Device Scope — What It Does and Does Not Do

The Device DOES	The Device Does NOT
Detect rhythm patterns from 6 known categories	Diagnose any specific medical condition
Flag unusual patterns that may warrant a doctor visit	Replace a cardiologist or clinical ECG
Operate fully offline with no cloud connectivity	Store or transmit patient data anywhere
Output plain-language messages anyone can understand	Show raw ECG waveforms or medical numbers to users
Run for hours on a rechargeable LiPo battery	Require a smartphone or internet connection

03 System Architecture & Signal Pipeline

End-to-End Signal Processing Pipeline

The device captures an electrical signal from the skin surface (the same signal measured in hospital ECGs), cleans it using an AI denoiser running locally on the chip, classifies the rhythm pattern against known cardiac conditions, and displays a plain-language result on a small screen. The entire process happens in under 100 milliseconds, with no data ever leaving the device.

Stage	Component	Function	Latency
1 — Capture	Skin electrodes + AD8232	Amplifies mV-level skin electrical signal; analog band-pass filters baseline drift and 50/60 Hz interference before ADC	Hardware
2 — Sampling	ESP32 ADC (12-bit)	Samples at 360 Hz; stores rolling 10-second window in SRAM as int16 array	~0 ms
3 — Denoising	LSTM denoiser (TFLite Micro)	Sequence-to-sequence network removes motion artifacts, EMI, muscle noise from each window	~30 ms
4 — Classification	1D-CNN classifier (TFLite Micro)	Identifies rhythm from 6 categories; outputs class + confidence score	~20 ms
5 — Interpretation	Rule-based C++ interpreter	Maps class + confidence to one of 4 plain-language outputs; flags uncertain results	< 1 ms
6 — Display	SSD1306 OLED (I2C)	Renders plain-language message; shows battery level indicator	< 5 ms

Rhythm Classification Categories

The classifier recognizes six rhythm categories, each mapped to a plain-language output and urgency tier:

Rhythm Category	Description	Urgency	User Message
Normal Sinus Rhythm	Regular rate, normal P-QRS-T morphology	None	Heart rhythm looks normal.
Atrial Fibrillation (AFib)	Irregular rhythm, absent P waves, variable R-R intervals	High	Irregular rhythm detected. Please seek medical attention.
Premature Ventricular Contraction	Early wide QRS complex, compensatory pause	Medium	Unusual rhythm detected. Please see a doctor soon.
Tachycardia	Heart rate > 100 bpm with consistent morphology	Medium	Unusual rhythm detected. Please see a doctor soon.
Bradycardia	Heart rate < 50 bpm with	Medium	Unusual rhythm detected.

Rhythm Category	Description	Urgency	User Message
	consistent morphology		Please see a doctor soon.
ST Abnormality	Elevated or depressed ST segment; ischemia marker	High	Irregular rhythm detected. Please seek medical attention.

PLAIN-LANGUAGE OUTPUT DESIGN

The interpreter never shows raw ECG data, confidence percentages, or medical terminology to the end user. The four possible outputs are: (1) Heart rhythm looks normal. (2) Unusual rhythm detected. Please see a doctor soon. (3) Irregular rhythm detected. Please seek medical attention. (4) Signal unclear. Please reattach electrodes and try again.

04 Hardware Stack & Bill of Materials

Component Overview

Every component was selected to minimize cost while maintaining clinical-grade signal quality sufficient for rhythm classification. The total bill of materials (BOM) for a single unit targets under \$15 USD, sourced from standard electronics suppliers (AliExpress, LCSC, Digi-Key).

Component	Model / Spec	Role	Unit Cost (USD)
Microcontroller	ESP32-WROOM-32 (or Dev board)	240 MHz dual-core, 520 KB RAM, 4 MB Flash. Runs all AI inference and firmware. Built-in BT/Wi-Fi (unused).	\$3.50
ECG Analog Frontend	AD8232 (SparkFun breakout)	Instrumentation amplifier + band-pass filter. Amplifies mV-level cardiac signals. Built-in right-leg-drive circuit.	\$2.00
OLED Display	SSD1306, 0.96", 128x64 px, I2C	Low-power monochrome display. Shows plain-language result. Visible in sunlight. No backlight drain.	\$1.50
Electrodes & Cable	Snap-type reusable cable + disposable gel pads	Adhesive patches placed on chest or wrists. Snap connector for reusable lead wire. Gel pads ~\$0.10 each.	\$1.50
Power — Battery	LiPo 500–1000 mAh (3.7V)	Portable rechargeable power. 500 mAh gives ~6 hours continuous use at 80 mA average draw.	\$1.50
Power — Charger	TP4056 USB-C charge module	LiPo charge controller with overcharge and over-discharge protection. Standard micro-USB or USB-C.	\$0.50
Passives	Resistors, capacitors, connectors	Decoupling caps on AD8232 supply rails; RC filter on ADC input; headers and JST connectors.	\$0.50
PCB	Custom 2-layer, 60x40 mm (JLCPCB)	Eliminates breadboard unreliability. \$2 for 5 boards from JLCPCB at minimum order.	\$0.40
TOTAL BOM (single unit)			~\$11.90

Circuit Description

Analog Signal Path

The AD8232 is a purpose-built integrated circuit for ECG signal conditioning. It contains a fully differential instrumentation amplifier with configurable gain, an operational amplifier for additional gain stages, a right-leg-drive (RLD) circuit to reduce common-mode interference, and an internal band-pass filter.

The signal path from skin to ADC is as follows: (1) Differential voltage from two electrodes enters the AD8232 IN+ and IN- pins. (2) The RLD electrode on the third lead (right leg or right arm) actively drives the common-mode voltage toward zero. (3) The AD8232 amplifies the

differential signal by approximately 100x and band-pass filters to 0.5–40 Hz (the clinically relevant ECG frequency range). (4) The OUTPUT pin connects to GPIO34 of the ESP32 (ADC1 channel). (5) The LOFF+ and LOFF- pins monitor electrode contact; firmware reads these to detect disconnected leads.

Digital Processing Path

The ESP32 ADC samples at 360 Hz (matching the MIT-BIH Arrhythmia Database native sample rate). A rolling circular buffer stores the most recent 1296 samples (3.6 seconds — approximately 3–5 heartbeats). The firmware runs a 10-second sliding window analysis loop, feeding 3600-sample windows into the TFLite Micro inference engine.

Power System

The TP4056 charge controller accepts USB 5V input and manages LiPo charging at up to 1A. A low-dropout 3.3V regulator (AMS1117-3.3) supplies the ESP32 and AD8232. A voltage divider on GPIO35 monitors the battery voltage. The firmware displays low-battery warning and disables inference at < 3.4V to prevent ADC noise from LiPo voltage sag corrupting readings.

05 Software Stack & AI Architecture

Two-Environment Architecture

The software exists in two separate environments: a training environment (Python on desktop/cloud GPU, used once) and a deployment environment (C++ on ESP32, runs permanently).

Layer	Environment	Technology	Purpose
Data Ingestion	Training (Python)	WFDB library, NumPy, SciPy	Parse MIT-BIH Arrhythmia Database (48 annotated 30-min recordings). Extract beat-labeled segments at 360 Hz. Apply augmentation and noise injection pipeline.
Noise Injection	Training (Python)	NumPy, custom pipeline	Add motion artifact, baseline wander, 50/60 Hz PLI, EMG at SNR levels from 0 dB to 40 dB. Creates 5x data augmentation for denoiser training.
Denoiser Model	Training + Deployment	Keras LSTM → TFLite Micro (int8)	Seq-to-seq LSTM: 2 layers, 64 hidden units, input window 360 samples. Trained to reconstruct clean ECG from noisy input. Post-training int8 quantization to 148 KB.
Classifier Model	Training + Deployment	Keras 1D-CNN → TFLite Micro (int8)	1D-CNN: 3 conv blocks (32/64/128 filters, kernel 5), global average pool, softmax head (6 classes). Trained on denoised segments. Quantized to 112 KB.
Quantization	Optimization (Python)	TensorFlow Lite post-training int8	Reduces model size ~4x and inference time ~3-4x. Full integer quantization with representative dataset calibration. Both models fit in ESP32 SRAM simultaneously.
Firmware	Deployment (C++)	Arduino framework + ESP-IDF	State machine: IDLE → SAMPLING → INFERENCE → DISPLAY → IDLE. FreeRTOS tasks: ADC sampling (Core 0), inference + display (Core 1). Deterministic real-time loop.
Interpreter	Deployment (C++)	Rule-based logic	Confidence threshold: > 0.75 triggers result display; 0.55–0.75 triggers 'signal unclear'; < 0.55 requests electrode reattachment. Maps class to urgency tier and message string.
Model Tools	Training (Python)	TensorFlow, Keras, sklearn	Training loop, early stopping, confusion matrix evaluation, ROC-AUC per class, McNemar test for statistical comparison vs. baseline.

Model Architecture Details

LSTM Denoiser

Architecture: Encoder-decoder LSTM. Input: 360-sample window (1 second at 360 Hz), normalized to $[-1, 1]$. Encoder: 2 LSTM layers (64 units each, `return_sequences=True`). Decoder: 2 LSTM layers (64 units) + Dense output layer (360 units, linear activation). Loss: MSE on clean ECG reconstruction. Training data: MIT-BIH recordings + synthetic noise at 8 SNR levels. Post-quantization size: ~148 KB.

1D-CNN Classifier

Architecture: Three convolutional blocks. Each block: Conv1D → BatchNorm → ReLU → MaxPool. Filter counts: 32 / 64 / 128. Kernel size: 5 across all layers. Global Average Pooling after final conv block. Dense(64, ReLU) → Dropout(0.5) → Dense(6, Softmax). Input: denoised 360-sample window. Output: 6-class probability vector. Post-quantization size: ~112 KB.

MEMORY BUDGET

ESP32 total SRAM: 520 KB. Allocation: Denoiser model: 148 KB. Classifier model: 112 KB. Sample buffer (3600 x int16): ~7 KB. FreeRTOS overhead: ~40 KB. Display buffer: ~1 KB. Available headroom: ~212 KB. Both models fit simultaneously with no flash overlay needed.

06 Circuit Schematic Overview

System Block Diagram

The following table represents the system interconnection schematic. A full KiCad schematic (.sch) and PCB layout (.kicad_pcb) will be produced in Sprint 1 and open-sourced with the project.

Block	Signals / Connections	Notes
Electrode LA (Left Arm)	AD8232 IN+	Positive differential ECG input
Electrode RA (Right Arm)	AD8232 IN-	Negative differential ECG input
Electrode RL (Right Leg)	AD8232 SDN/RLD	Right-leg drive — common-mode rejection
AD8232 OUTPUT	ESP32 GPIO34 (ADC1_CH6)	Filtered, amplified ECG signal ~0.5–2.5 V
AD8232 LOFF+	ESP32 GPIO32	Lead-off detection positive
AD8232 LOFF-	ESP32 GPIO33	Lead-off detection negative
AD8232 VCC	3.3 V rail	100 nF decoupling cap to GND at pin
SSD1306 SDA	ESP32 GPIO21 (I2C SDA)	Display data line — 4.7 kOhm pull-up to 3.3 V
SSD1306 SCL	ESP32 GPIO22 (I2C SCL)	Display clock line — 4.7 kOhm pull-up to 3.3 V
SSD1306 VCC	3.3 V rail	100 nF decoupling cap
TP4056 VOUT+	LiPo BAT+ → AMS1117 VIN	4.2 V max LiPo output
AMS1117-3.3 VOUT	ESP32 3V3, AD8232 VCC, SSD1306 VCC	Regulated 3.3 V supply to all digital components
LiPo BAT+ (via divider)	ESP32 GPIO35 (ADC1_CH7)	10k/10k divider — battery voltage monitor
USB-C connector VBUS	TP4056 VIN	5 V USB charging input
ESP32 EN button	ESP32 EN pin → 10k pull-up to 3.3 V	Hardware reset button

PCB NOTES

Target PCB dimensions: 60 x 40 mm (fits inside a standard project enclosure). 2-layer stackup. Signal traces on Top copper; power and ground pour on Bottom copper. Electrode connector: 3-pin 2.54 mm JST-PH. Battery connector: 2-pin 2.0 mm JST-PH. Display connector: 4-pin 2.54 mm header. Manufacturing: JLCPCB (5 boards, ~\$10 total including shipping).

07 Cost Analysis & Budget

Bill of Materials — Detailed Cost

Item	Quantity	Unit Cost (USD)	Total (USD)	Supplier
ESP32-WROOM-32D module	3	\$2.80	\$8.40	LCSC / AliExpress
AD8232 ECG breakout board	3	\$1.80	\$5.40	AliExpress
SSD1306 OLED 0.96" I2C	3	\$1.20	\$3.60	AliExpress
LiPo battery 800 mAh 3.7V	3	\$1.50	\$4.50	AliExpress
TP4056 USB-C charge module	3	\$0.40	\$1.20	AliExpress
AMS1117-3.3 SOT-223	10	\$0.10	\$1.00	LCSC
Electrode snap cable (3-lead)	3	\$1.00	\$3.00	AliExpress
Disposable gel electrode pads	50	\$0.10	\$5.00	Amazon
Resistors assortment (SMD 0402)	1 kit	—	\$3.00	LCSC
Capacitors assortment (SMD 0402)	1 kit	—	\$3.00	LCSC
JST connectors (PH + ZH)	10 sets	—	\$2.00	AliExpress
Custom PCB (JLCPCB, 5 boards)	1 order	—	\$9.50	JLCPCB
Dev board headers + misc.	—	—	\$3.00	Local
		SUBTOTAL (3 prototypes)	\$53.60	
		Contingency (+15%)	\$8.04	
		TOTAL BUDGET	\$61.64	
		Cost per prototype unit	~\$20.55	(includes shared tooling)

Development & Non-Hardware Costs

Item	Cost	Notes
GPU compute — model training	\$0	Google Colab free tier (T4 GPU, estimated 4–8 hours total)
MIT-BIH Arrhythmia Database	\$0	Freely available on PhysioNet (open access)
Software toolchain	\$0	All open-source: TensorFlow, Arduino IDE, KiCad, Python
MDPI Sensors publication APC	~\$2,200	Article Processing Charge if accepted (fee waiver may apply for student authors)
IRB / ethics review (feasibility study)	~\$0–\$50	University ethics board — typically waived for low-risk observational studies
Miscellaneous (shipping, tools)	~\$30	Multimeter, oscilloscope probe, soldering consumables

Cost Comparison — Market Context

Product	Price	Connectivity Required	Offline AI	Target User
HeartLens AI (this project)	~\$12 BOM	None	Yes (on-device)	Underserved / rural / no smartphone
AliveCor KardiaMobile	\$99 + \$99/yr sub	Smartphone required	No (cloud)	Smartphone users, urban
Withings ScanWatch	\$299	Smartphone required	No (cloud)	Affluent tech consumers
Clinical ECG (hospital)	\$50–\$200/session	Hospital visit	N/A	Patients with insurance
Holter Monitor (24-hr rental)	\$200–\$500	Hospital infrastructure	N/A	High-risk cardiac patients

08 Project Timeline — 16-Week Gantt Chart

Sprint Overview

The project is organized into 5 sprints across 16 weeks. Each sprint has a defined deliverable milestone. Work is part-time (~10–15 hours per week alongside university coursework).

Sprint	Weeks	Theme	Key Deliverable
Sprint 1	W1–W3	Hardware & Signal Acquisition	Prototype board assembled; clean ECG signal verified on oscilloscope
Sprint 2	W4–W7	Data Pipeline & Model Training	Both AI models trained, quantized, and benchmarked on MIT-BIH
Sprint 3	W8–W10	Firmware Integration	Full pipeline running on ESP32; latency and RAM measurements recorded
Sprint 4	W11–W13	Noise Study & Feasibility Testing	Real-world noise study complete; small participant study conducted
Sprint 5	W14–W16	Documentation & Publication	Paper draft submitted; all hardware/code open-sourced on GitHub

Detailed Week-by-Week Gantt

Task	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13	W 14	W 15	W 16
KiCad Schematic + PCB layout	■	■														
PCB order + component sourcing		■	■													
PCB assembly + AD8232 bringup			■													
Signal quality verification (scope)			■													
MIT-BIH data download + WFDB parse				■												
Noise injection pipeline				■	■											
Denoiser LSTM training					■	■										
CNN classifier training						■	■									
int8 quantization + size check							■									
ADC sampling firmware								■								

Task	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13	W 14	W 15	W 16
(FreeRTOS)								■								
TFLite Micro integration								■	■							
Interpreter + display firmware								■								
Full pipeline integration + latency test								■	■							
Real-world noise capture study										■	■					
Participant feasibility study (n=10)											■	■				
Confusion matrix + per-class analysis												■				
Paper draft (methods + results)													■	■		
Open-source release (GitHub)														■	■	
Paper submission + revision															■	■

MILESTONE GATES

Sprint 1 exit gate: Clean ECG visible on oscilloscope (SNR > 20 dB, baseline stable).
 Sprint 2 exit gate: Classifier achieves F1 > 0.87 on clean MIT-BIH test set. Sprint 3 exit gate: Full pipeline latency < 100 ms on hardware. Sprint 4 exit gate: Accuracy on real-world capture within 8% of clean benchmark. Sprint 5 exit gate: Paper submitted to Sensors or IEEE IoT Journal.

09 Research Contribution & Publication Strategy

Three Distinct Contributions

1. Full-Stack Edge Inference Within Hard Hardware Constraints

Prior work typically benchmarks models on desktop hardware and claims 'deployability.' This project implements and measures the complete pipeline — denoise, classify, interpret — running simultaneously on a \$3.50 microcontroller, with real latency and RAM measurements published.

2. Systematic Real-World Noise Robustness Study

The noise injection pipeline (motion artifact, baseline wander, 50/60 Hz interference, muscle noise at varying SNR levels) is compared against ECG captured from the actual prototype in real conditions. This synthetic vs. real-world noise comparison is underrepresented in embedded ECG literature.

3. Deployment-Framed Feasibility Study

A small study with real participants produces honest end-to-end accuracy benchmarks — not 'accuracy on a clean benchmark dataset' but 'accuracy when a real person attaches electrodes in a non-clinical setting.' This is the number that actually matters for real-world deployment.

Target Venue	Type	Scope Fit	Impact Factor
Sensors (MDPI)	Primary — Journal	IoT + biomedical sensing, edge AI, open hardware	Q2, IF ~3.9
IEEE Internet of Things Journal	Alternate — Journal	Edge computing, embedded systems, healthcare IoT	Q1, IF ~10.6
Biomedical Signal Processing & Control	Alternate — Journal	ECG signal processing, clinical biomedical engineering	Q1, IF ~5.1
IEEE EMBC Conference	Conference (backup)	Biomedical engineering — poster or short paper track	Top-tier in field

REGULATORY SCOPE — EXPLICITLY STATED IN THE PAPER

HeartLens AI is a preventative screening aid, not a medical device. It is not intended to diagnose, treat, or manage any condition. Its sole function is to flag rhythm patterns that may warrant professional evaluation. All performance claims will be scoped accordingly. This framing is not a limitation — it is the correct and honest characterization of what the device does, and reviewers in this space respond well to authors who demonstrate they understand the regulatory distinction.

10 Risk Register & Mitigation

Risk	Likelihood	Impact	Mitigation
SRAM overflow: both models exceed 520 KB	Medium	High	Pre-calculated: 260 KB total + OS overhead fits. Fallback: use flash overlay (esp_partition_mmap) to page denoiser in/out. Last resort: switch to lighter MobileNet-style architecture.
AD8232 signal quality insufficient in field	Medium	High	Test on oscilloscope at Sprint 1 gate. If SNR < 15 dB, add external instrumentation amp stage (INA333). Spiral electrode pattern as backup for wrist capture.
MIT-BIH class imbalance (AFib << Normal)	High	Medium	Apply SMOTE oversampling and class-weighted loss during training. Validate per-class F1, not just accuracy. Target F1 > 0.85 on minority classes.
Participant recruitment for feasibility study	Low–Medium	Medium	University ethics board approval first. Target 10 healthy volunteers (colleagues, family). Expand to 20 if time allows. Study is observational — minimal IRB complexity.
Quantization accuracy degradation > 5%	Medium	Medium	Use representative calibration dataset (500 samples per class). If degradation > 5% on any class, apply quantization-aware training (QAT) in Keras before re-exporting.
PCB manufacturing defect / component failure	Low	Medium	Order 5 PCBs (JLCPCB minimum order), 3 of each component. Breadboard fallback available for all Sprint 2 work. Keep 2 spare ESP32 boards.
Scope creep delaying publication	High	Medium	Strict sprint gates. Scope is fixed: no new rhythm classes, no BLE feature, no app development. Gantt reviewed weekly.

07 — Conclusion

Why This Project Matters

The technology to build HeartLens AI exists today, is inexpensive, is well-understood, and is sitting unused in the gap between clinical-grade devices and the billions of people who will never have regular access to them.

This project does not require a breakthrough. It requires careful engineering, honest evaluation, and a willingness to build for the actual user — not the user who already has a smartphone, a cardiologist, and a stable internet connection.

The hardware costs ~\$12. The software runs entirely offline. The output is four sentences that anyone can understand.

"The gap between detectable and detected is not a scientific problem. It is an access problem. HeartLens AI is one small attempt to close it."